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ACROSS TIME AND SPACE

VISUALIZING ENVIRONMENTAL CHANGE FROM MULTI-TEMPORAL SATELLITE IMAGES

01 PIPELINE

READING IMAGE

- Sort multi-temporal images by time
- Call multi-image analysis (>2)



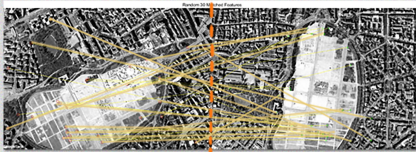
PREPROCESSING

- Pairwise comparison with the first image
- Grayscale adjustment



MATCHING

- detectSURFfeatures: feature matching
- estgeform2d: mapping relationship
- imwarp: image warping



RENDERING

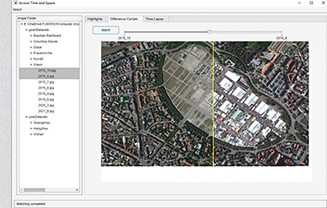
- Align with the first image



02 GUI

THREE VIEWING MODES

- Three viewing modes
- Highlights
- Image Curtain
- Time Lapse



03 PROBLEMS

- Fake Matches
- Algorithm selection
- Image comparison order issues
- Global registration failure
- Rendering issues

04 METHODS

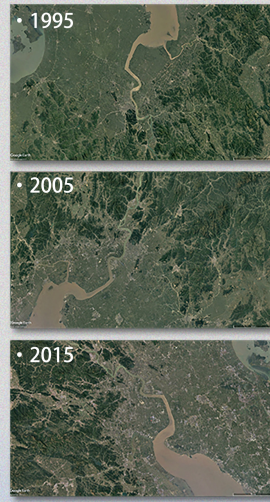
- Adjust parameter thresholds
- Use stable and efficient SURF library
- Always compare with the first image
- Image subdivision into 4 or 9 parts
- Fill black regions

05 RESULTS

Hangzhou, China

ORIGINAL IMAGE

- 1995
- 2005
- 2015



OUTPUT IMAGE

- 1995
- 2005
- 2015



THE 1ST IMAGE

- ORIGINAL IMAGE
- STANDARD IMAGE
- OUTPUT IMAGE

